

Problem

In long-context LLMs, evicting parts of the KV-cache can either **HELP** or **HURT**

HELPS (dilute-prone tasks):



- Many redundant prefill tokens dilute attention, removing them improves accuracy

HURTS (capacity-bound tasks):

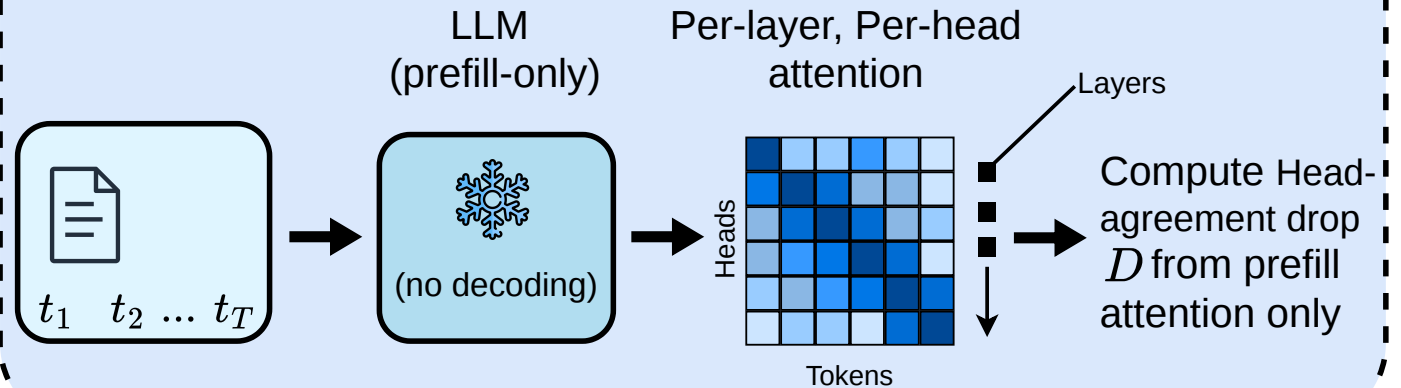


- Answer depend on precise retrieval of a few specific tokens
- Eviction causes significant failure

Can we predict, before decoding, whether eviction will **help** or **hurt** for a given output?



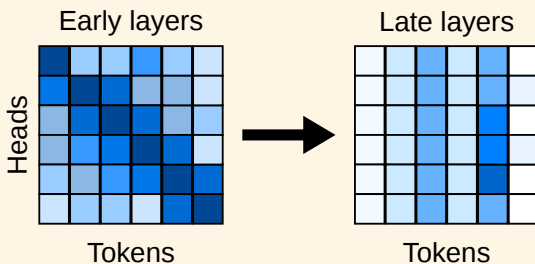
Setup: Prefill → Attention Analysis



Intuition: Why D Works?

Dilution-prone (large D)

Early layers: High head agreement over redundant tokens
Late layers: Heads specialize → agreement decreases.

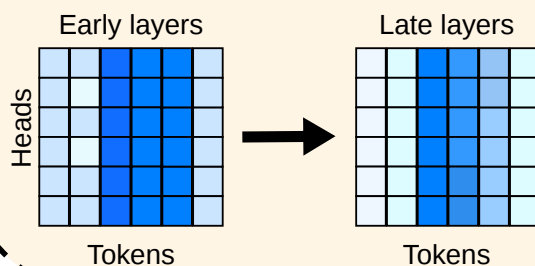


Eviction removes redundancy → attention concentrates on important tokens → helps



Capacity-bound (small D)

Heads already agree early and stay agreed



Eviction likely drops few critical tokens → accuracy collapses



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